Expected Fewest Moves to Solve a Scrambled Cube: The $3 \times 3 \times 3$ Case and the $n \times n \times n$ Asymptotics

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Abstract

In the World Cube Association's *Fewest Moves Challenge* (FMC) a competitor is handed one scramble and must write down the shortest solution they can find in 60 minutes, scored by move count. This note asks the *information-theoretic floor* of that event: given a random (or WCA) scramble, what is the expected *optimal* solution length — the fewest moves that exist, which no competitor can beat? For the $3\times3\times3$ this is the expected value of the "God's-number distance" of a random state; using the exact/near-exact distance distribution behind the proof that God's Number is 20, we get

$$\mathbb{E}[\text{optimal HTM moves}] \approx 17.70,$$

with over 99.9% of states at distance 17 or 18. We state the result in both the half-turn metric (HTM, the usual theoretical metric) and note the WCA's official OBTM scoring. For the general $n \times n \times n$ cube we invoke the theorem of Demaine, Demaine, Eisenstat, Lubiw and Winslow (2011) that the diameter — and, by a counting argument, the *expected* optimal length of a random scramble — is $\Theta(n^2/\log n)$. We make the counting lower bound explicit: $\mathbb{E}[\text{optimal moves}] = \Omega(n^2/\log n)$, and combine it with their $O(n^2/\log n)$ algorithm to conclude $\mathbb{E} = \Theta(n^2/\log n)$. A short table gives the constant-free growth for $n = 2, 3, 4, 5$.

1 The event, precisely

Definition 1 (WCA Fewest Moves Challenge)**.** In FMC a single scramble is given to all competitors on paper; each has 60 minutes to record a solution. The result is the move count (lower is better), averaged over three attempts (mean of 3; any DNF makes the mean a DNF). A solution must be at most 80 moves. The official metric is **OBTM** (Outer Block Turn Metric): single-layer and wide (block) turns each count 1, inner slice moves are not used in notation, and whole-cube rotations are free [1]. Scrambles since 2016 begin and end with `R' U' F` so that solutions cannot be read off the scramble.

A *competitor* finds some solution; the *event's floor* is the *optimal* (shortest possible) solution, the quantity a perfect solver would submit. We study the floor, because (i) it is well-defined and metric-dependent only, and (ii) it bounds every competitor from below. We work in the half-turn metric (HTM: any single face turn 90° or 180° counts 1), the metric in which the distance distribution is known; §3 relates it to OBTM.

Definition 2 (Optimal length / God's distance)**.** For a state s of the cube group G, let $d(s)$ be the minimum number of (HTM) moves solving s. The *expected fewest moves* for a random scramble is $\mathbb{E}_{s\sim\mathrm{Unif}(G)}[d(s)]$. The *diameter* is $\max_s d(s)$ ("God's Number").

2 The $3 \times 3 \times 3$: expected optimal length ≈ 17.70

For the $3 \times 3 \times 3$, $|G| = 4.3 \times 10^{19}$ and the diameter is exactly 20 in HTM (Rokicki–Kociemba–Davidson–Dethridge, 2010 [2]). The full distance distribution is known exactly up to distance 15 and by tight estimate for 16–20 [2, 3]:

distance k	number of states N_k	fraction $N_k/\lvert G\rvert$
≤ 14	$\approx 1.2 \times 10^{16}$	$\approx 0.00003\%$
15	91,365,146,187,124,313	0.21%
16	$\approx 1.1 \times 10^{18}$	2.6%
17	$\approx 1.2 \times 10^{19}$	26.7%
18	$\approx 2.9 \times 10^{19}$	67.0%
19	$\approx 1.5 \times 10^{18}$	3.4%
20	$\approx 4.9 \times 10^{8}$	$\approx 10^{-11}$

Table 1: HTM optimal-distance distribution for the $3 \times 3 \times 3$ [2, 3]. Values for $k \geq 16$ are tight estimates; $k \leq 15$ are exact. Fractions are the standard reported figures.

Theorem 1 (Expected fewest moves, $3 \times 3 \times 3$)**.** *Under the uniform distribution on G,*

$$\mathbb{E}[d(s)] \;=\; \sum_{k} k\,\frac{N_k}{|G|} \;\approx\; 17.70 \text{ HTM moves},$$

and the distribution is extremely concentrated: $\Pr[d \in \{17, 18\}] \approx 93.7\%$ *and* $\Pr[d \in \{16, 17, 18, 19\}] \approx 99.7\%$.

Proof. By definition of expectation over the finite set G, $\mathbb{E}[d] = \sum_k k\,(N_k/|G|)$. Substituting the fractions of Table 1,

$$\mathbb{E}[d] \approx 15(0.0021) + 16(0.026) + 17(0.267) + 18(0.670) + 19(0.034) + 20(5\times10^{-12}) \approx 17.70,$$

matching the independently reported average optimal length ≈ 17.7 obtained by Kociemba from optimally solving 100,000 random states [3]. The concentration figures are the sums of the relevant rows. □

Remark 1 (What this says about the event)**.** No competitor, however skilled, can average below ≈ 17.7 HTM moves against random scrambles — that is the mathematical floor of 3x3 FMC. In practice humans in a 60-minute window average in the high-20s, and the best single solves reach into the low-20s; the gap to ≈ 17.7 is the "human/search" overhead above the optimum. (World-class single FMC results of 16–20 moves occur on scrambles whose optimum is itself low and found by extensive insertion search.)

3 Metric note: HTM vs. OBTM

On the $3 \times 3 \times 3$ the WCA scores OBTM, which for a $3 \times 3 \times 3$ assigns cost 1 to each outer face quarter- or half-turn — identical to HTM for the $3\times3\times3$ since no wide/slice moves shorten a $3\times3\times3$ solution's count beyond face turns (wide moves equal a face turn plus a rotation, and rotations are free). Thus the ≈ 17.70 figure is the OBTM floor as well. The metrics genuinely diverge only for $n \geq 4$, where wide (block) turns move many layers at once and are individually cheaper than reproducing their effect with single-layer turns — which is exactly where the asymptotic savings below come from.

4 The $n \times n \times n$: expected fewest moves is $\Theta(n^2/\log n)$

For general n the exact distribution is hopeless, but the *growth rate* is settled.

Theorem 2 (Demaine–Demaine–Eisenstat–Lubiw–Winslow, 2011 [4])**.** *The diameter (God's Number) of the $n \times n \times n$ Rubik's cube group, in the outer-block/face metric, is $\Theta\left(\dfrac{n^2}{\log n}\right)$.*

The upper bound comes from parallelizing the standard $\Theta(n^2)$ layer-by-layer solve: many commuting cubie-moves can be batched into one turn, saving a $\log n$ factor. The lower bound is a counting (information-theoretic) argument, which we restate because it controls the *expected*, not just worst-case, length.

Lemma 1 (Counting lower bound, in expectation)**.** *Let $M = M(n)$ be the number of distinct single moves available on the $n \times n \times n$ cube. Then $M = \Theta(n)$ (each of the 3 axes has $\Theta(n)$ movable layers, each with 3 nontrivial turn amounts). The number of reachable states satisfies $\log |G_n| = \Theta(n^2)$ (there are $\Theta(n^2)$ facelets/cubies carrying $\Theta(n^2)$ "bits" of configuration). If every state had an optimal solution of length $\le L$, then every state is a product of $\le L$ moves, so*

$$|G_n| \;\le\; \sum_{\ell=0}^{L} M^{\ell} \;\le\; (M+1)^{L}.$$

Taking logs, $L \;\ge\; \dfrac{\log |G_n|}{\log(M+1)} = \dfrac{\Theta(n^2)}{\Theta(\log n)} = \Omega\left(\dfrac{n^2}{\log n}\right)$. Because at most a $(M+1)^{L'}/|G_n|$ fraction of states have distance $\le L'$, for any $L' = o(n^2/\log n)$ that fraction is $|G_n|^{-\Omega(1)} \to 0$; hence all but a vanishing fraction of states have distance $\Omega(n^2/\log n)$, and therefore

$$\mathbb{E}_s[d_n(s)] \;=\; \Omega\left(\frac{n^2}{\log n}\right).$$

Proof. The sphere-size bound $|G_n| \le (M+1)^L$ is the standard counting bound for Cayley graphs: there are at most $M+1$ choices (including "stop") at each of L steps. Rearranging gives the worst-case L. For the average, note the number of states within distance L' is at most $(M+1)^{L'}$; dividing by $|G_n| = 2^{\Theta(n^2)}$, if $L' \log(M+1) \le (1-\varepsilon)\log|G_n|$ then this fraction is at most $|G_n|^{-\varepsilon} \to 0$. So a $1 - o(1)$ fraction of states lie at distance $\ge (1-\varepsilon)\log|G_n|/\log(M+1) = \Omega(n^2/\log n)$, and the expectation inherits the same lower bound. The matching upper bound $\mathbb{E}[d_n] \le \text{diam} = O(n^2/\log n)$ is Theorem 2. □

Theorem 3 (Expected fewest moves, general n)**.** *For the $n \times n \times n$ cube under a uniform random scramble,*

$$\boxed{\mathbb{E}[\textit{optimal moves}] \;=\; \Theta\left(\frac{n^2}{\log n}\right).}$$

In particular it grows essentially quadratically in n (with a $\log n$ discount), reflecting that the configuration has $\Theta(n^2)$ independent degrees of freedom while each move only carries $O(\log n)$ bits of "correction" on average.

Proof. Combine the lower bound $\mathbb{E}[d_n] = \Omega(n^2/\log n)$ (Lemma 1) with the upper bound $\mathbb{E}[d_n] \le \text{diam} = O(n^2/\log n)$ (Theorem 2). □

n	cube	expected fewest moves (floor), growth $\sim n^2/\log n$
2	Pocket Cube	exact diameter 11 HTM / 14 QTM, by full search of all 3,674,160 states [5, 6]
3	Rubik's Cube	$\mathbb{E} \approx 17.70$ HTM (Thm. 1); diameter 20
4	Rubik's Revenge	$\Theta(n^2/\log n)$; no exact mean known (state space $\sim 7.4 \times 10^{45}$)
5	Professor's Cube	$\Theta(n^2/\log n)$; $\sim 2.8 \times 10^{74}$ states

Table 2: The exact mean is known only for small n (full or near-full search); for $n \geq 4$ only the $\Theta(n^2/\log n)$ growth rate is proven.

5 Discussion and honest scope

- $3 \times 3 \times 3$ **mean (17.70).** The distance *fractions* for $k \geq 16$ are tight community estimates, not exact counts (only $k \leq 15$ is exact); the mean ≈ 17.70 is corroborated by Kociemba's direct optimal solve of 100,000 random states, so we quote it as "$\approx$," not an exact rational.
- **General n.** Theorem 3 is a growth rate: it pins the exponent but hides constants. We do *not* claim a numerical mean for $n \geq 4$; computing even the diameter for $n = 4$ is open.
- **Floor vs. human.** "Expected fewest moves" is the optimum a perfect solver attains, i.e. the floor of the event. Real FMC results sit well above it (search is hard in 60 minutes); the floor is the mathematically clean target this paper computes.
- **Random vs. WCA scramble.** WCA scrambles are (near-)uniform random states, so the uniform-G expectation applies to both; the `R' U' F` wrapper does not bias the optimal distance.

References

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[3] H. Kociemba, *God's Number is 20*, kociemba.org/moves20.htm (distance distribution; average optimal length ≈ 17.70 from 100,000 random optimal solves). Accessed 2026.

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[6] *God's Algorithm for the $2 \times 2 \times 2$ Pocket Cube*, Cube Lovers mailing-list archive (MIT AI Lab), 1981 — the original exhaustive computation establishing the 11/14 diameters.